Rainfall, Temperature and Mortality in the Philippines

Municipality-level response functions, 2006–2024

September 8, 2026

Data

The panel

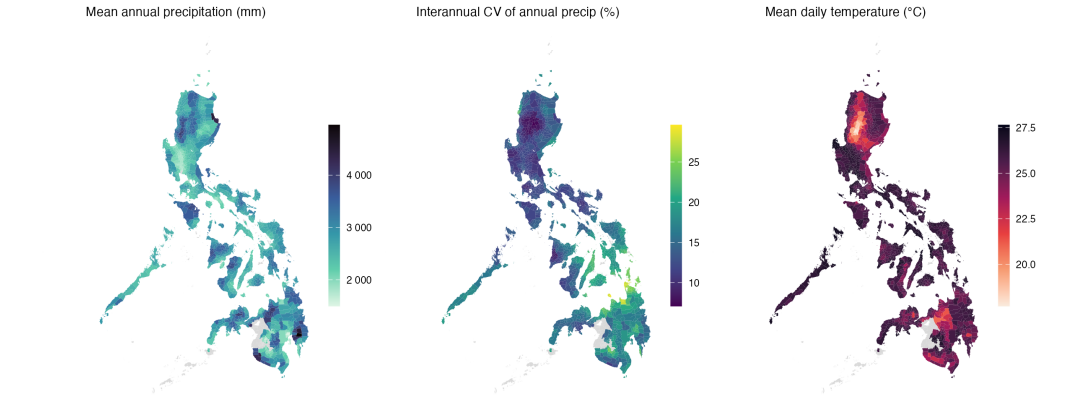
- New Philippines mortality panel, merged to gridded climate and GHSL settlement data.
- **1,475** municipalities (adm3), nested in **110** provinces (adm2) and **17** regions (adm1).
- Annual, 2006–2024, balanced. The pandemic years are retained and handled with a control rather than dropped (see the COVID section).
- Three age groups: 0–4, 5–64, 65+. The all-ages total (`agegroup` 0) is dropped.
- **84,075** municipality $\times$ year $\times$ age observations in the estimation sample. BARMM is almost entirely absent (8 of 127 municipalities) — see the maps.

Climate. Daily ERA5-Land precipitation and temperature, aggregated to annual sums of daily polynomial terms. CHIRPS precipitation is used as an alternative product.

Two notes on the climate variables

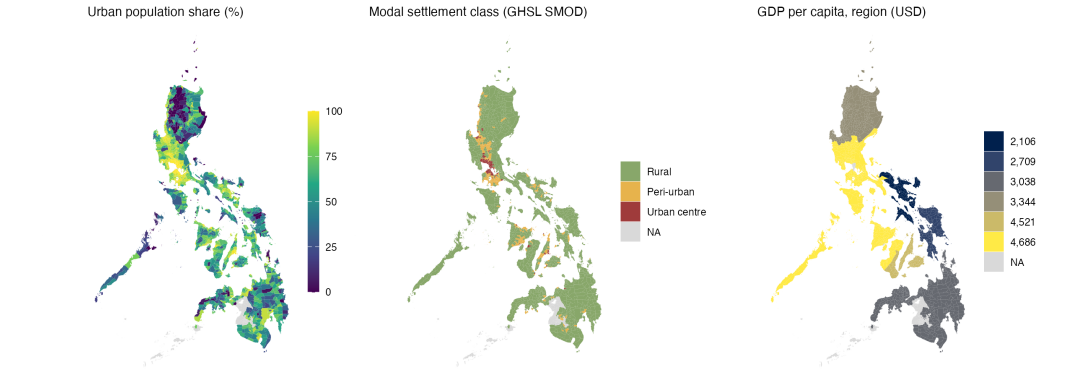
- The regressors are *annual sums of daily powers*, $\widetilde{P}_{it}^{(k)} = \sum_{d \in t} P_{id}^{k}$, not powers of the annual total. Response functions are therefore interpreted over **daily** values.
- Precipitation curves are evaluated over 0–300 mm/day and temperature over 15–35°C, which spans the support observed in the Philippines.

The climate



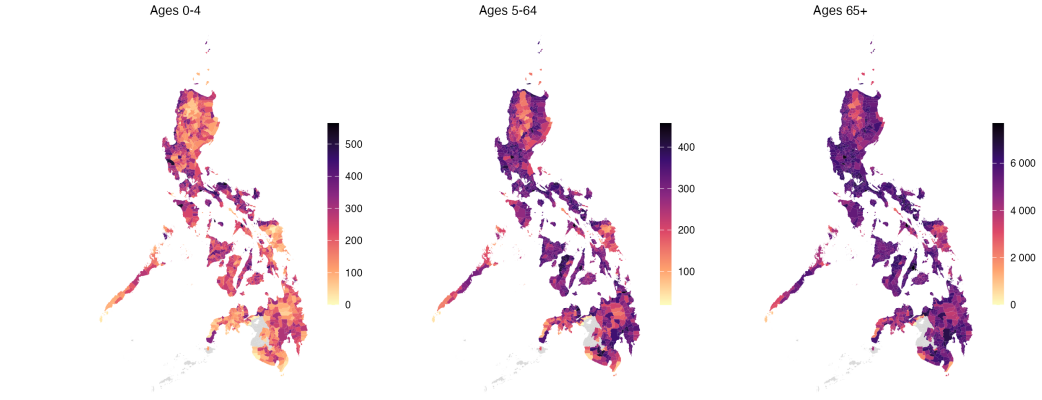
Municipality means over 2006–2024, ERA5-Land. The eastern seaboard — the typhoon-facing coast — is both the wettest and, in eastern Mindanao, the most variable year to year. Grey: municipalities absent from the mortality panel — almost all of BARMM (119 of its 127 municipalities; only 8 appear in the panel), reflecting civil-registration coverage there rather than a merge failure.

The heterogeneity covariates



Urban share is the GHSL population-based measure averaged over the sample; settlement class is each municipality's modal GHSL SMOD class; GDP per capita varies only at the region (adm1) level — the visual reminder of why the income margin is coarse.

Mortality geography



Mean winsorised death rates by municipality. Note the independent colour scales: 65+ rates are roughly twenty times the other groups. The sparse-looking north-Luzon interior for 0–4 reflects small cohorts and many zero-death municipality-years, the same cells the count Poisson and log-OLS estimators treat differently.

Summary Statistics

Summary statistics

Panel A: Mortality rate (deaths per 100,000)						
Age group	N	Mean	Std. dev.	Median	p99	Share zero
0-4	28,025	243.7	150.2	232.3	664.5	0.048
5-64	28,025	297.7	92.3	307.7	497.6	0.000
65+	28,025	5,339.6	1,387.3	5,456.3	8,814.5	0.002
Panel B: Climate						
Variable		Mean		Std. dev.	Min	Max
Annual precipitation, ERA5-Land (mm)		2,738.6		641.5	987.9	5,987.7
Annual precipitation, CHIRPS (mm)		2,800.4		846.0	785.3	7,064.5
Mean daily temperature, ERA5-Land (°C)		25.7		1.4	17.4	28.3
Long-run mean precipitation, adm1 (mm)		2,695.8		286.6	2,096.8	3,359.3
Long-run mean temperature, adm1 (°C)		25.8		1.0	22.0	26.9
Panel C: Heterogeneity margins						
Urban population share (%)		63.3		27.7	0.0	100.0
GDP per capita, adm1 average (USD)		3,650.3		880.4	2,105.9	4,686.0

Estimation sample. Mortality rates are deaths per 100,000. Climate variables are annual totals except temperature, reported as the implied daily mean.

Empirical Strategy

Estimating equation

For municipality i, age group a, year t:

$$D_{ait} = \exp\Big(\sum_{k=1}^{K_T} \beta_{ak}^{T} \, \widetilde{T}_{it}^{(k)} + \sum_{k=1}^{K_P} \beta_{ak}^{P} \, \widetilde{P}_{it}^{(k)} + \mu_{aic(t)} + \nu_{atr(i)} \Big) \cdot \varepsilon_{ait}$$

where

- D_{ait} is the death rate per 100,000, winsorised at the 99th percentile within age group;
- $\widetilde{X}_{it}^{(k)} = \sum_{d \in t} X_{id}^{k}$ is the annual sum of daily values raised to the kth power;
- $\mu_{aic(t)}$ is a municipality $\times$ age $\times$ pandemic-era effect, where $c(t)$ is the era containing year t (pre-2020, 2020–22, 2023–24);
- $\nu_{atr(i)}$ is a province $\times$ year $\times$ age effect, where $r(i)$ is the province containing municipality i.

Default uses $K_P = K_T = 3$ and specification (4) below. Estimated by Poisson pseudo-maximum likelihood, weighted by population share, with standard errors clustered at the province (adm2) level.

Response functions

The reported curve is the change in the death rate from a day at x relative to a reference day x_{ref}:

$$f_a^P(p) \;=\; \sum_{k=1}^{K_P} \hat{\beta}_{ak}^P \left(p^k - p_{\text{ref}}^k\right), \qquad p_{\text{ref}} = 0 \text{ mm}$$

$$f_a^T(\tau) \;=\; \sum_{k=1}^{K_T} \hat{\beta}_{ak}^T \left(\tau^k - \tau_{\text{ref}}^k\right), \qquad \tau_{\text{ref}} = 27^\circ\text{C}$$

Confidence intervals are delta-method, using the clustered variance matrix.

Heterogeneity. Each margin replaces β_{ak}^P with β_{akg}^P, where $g = g(i,t)$ indexes the heterogeneity cell, so the precipitation polynomial is fully interacted with age $\times$ group:

$$\sum_{k=1}^{K_P} \sum_{g} \beta_{akg}^P \, \mathbf{1}\{g(i,t) = g\} \, \widetilde{P}_{it}^{(k)}$$

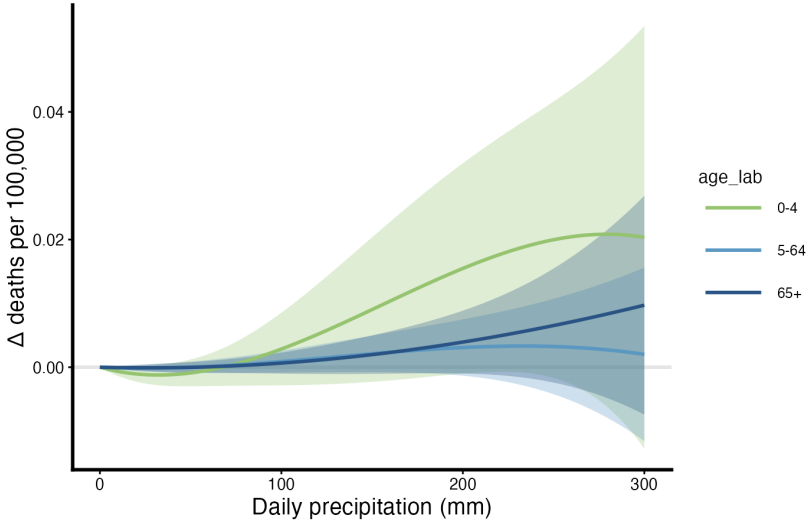
Fixed-effects specifications

	Fixed effects
(1) Year × age	$\mu_{ai} + \nu_{at}$
(2) + province-age trends	$\mu_{ai} + \nu_{at} + \delta_{ar(i)} \cdot t$
(3) Saturated	$\mu_{ai} + \nu_{atr(i)}$
(4) + pandemic control	$\mu_{aic(t)} + \nu_{atr(i)}$

μ_{ai} is a municipality × age effect; $r(i)$ denotes the **province** (adm2) containing municipality i and $c(t)$ the pandemic era containing year t. **(4) is the default** used throughout. These are the single-country analogues of the specifications used in the global Carleton-style regressions elsewhere in the project: with one country, country × year × age collapses to year × age.

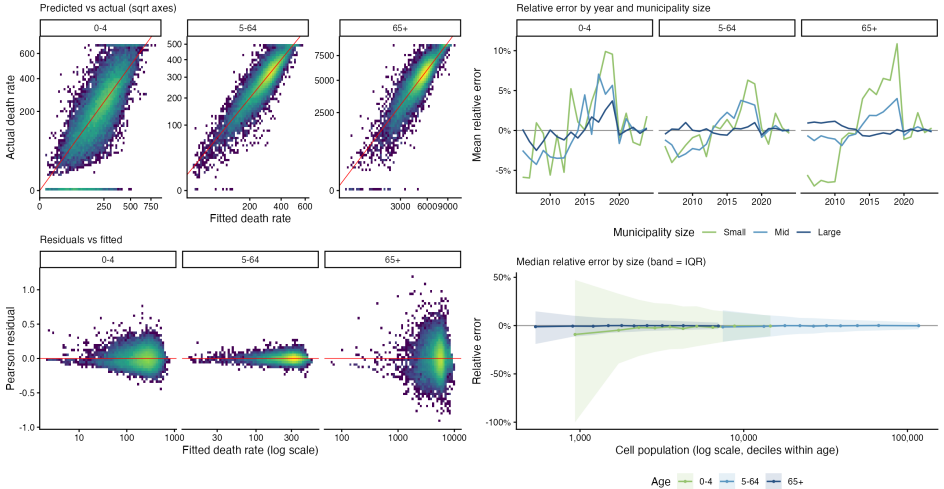
Baseline Response Functions

Precipitation response, by age group



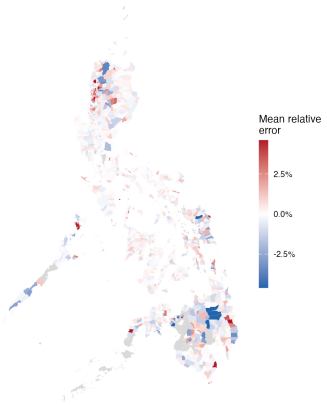
Default specification (4), full 2006–2024 panel with the pandemic control. Change in deaths per 100,000 relative to a day with no rainfall. Shaded bands are 95% confidence intervals.

Fit diagnostics for the default specification



Non-lean refit of the default model; squared correlation between actual and fitted is 0.62 / 0.82 / 0.79 for ages 0–4 / 5–64 / 65+, almost all of it from the fixed effects rather than from climate. The right-hand panels use relative error $(y - \hat{\mu})/\hat{\mu}$, which is scale-free and unaffected by the weighting; population bins are formed *within* age group.

Residual geography

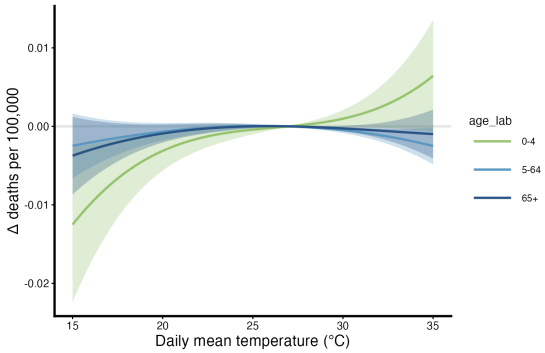


Mean relative error $(y - \hat{\mu})/\hat{\mu}$ by municipality, pooled over years and ages; the colour scale is clipped at the 99th percentile of |error|.

The pattern is close to spatial noise — no coherent typhoon-corridor or coastal signature, and no island-group block. That is reassuring for the fixed-effects design: what the model misses looks idiosyncratic rather than geographic.

Grey municipalities are those absent from the panel, overwhelmingly BARMM.

Temperature response — not identified in this panel



The fitted curves are wrong-signed somewhere at every age: hotter days imply *lower* mortality for 5–64 (significantly) and 65+, while for 0–4 the heat slope is positive but *cold* days imply significantly lower mortality than a 27°C day.

Not a solver problem (diagnostics converge in a handful of iterations; tighter tolerances move coefficients by $\sim 3 \times 10^{-12}$) — an identification problem:

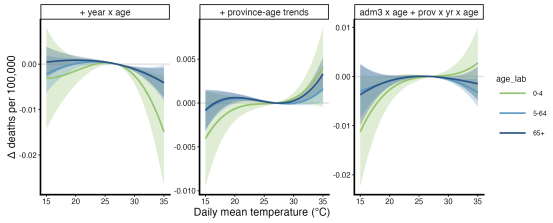
- the curves reshape substantially across fixed-effects sets (next slide);
- the polynomial terms are 0.99 collinear after the fixed effects.

Temperature is retained as a control, not interpreted.

Why: the fixed effects absorb the temperature signal

	Precipitation	Temperature
% of s.d. surviving FEs	49.3	9.1
corr(linear, quad.) post-FE	0.789	0.991
variance inflation factor	2.6	54.8
residual daily variation	0.87 mm	0.127 °C

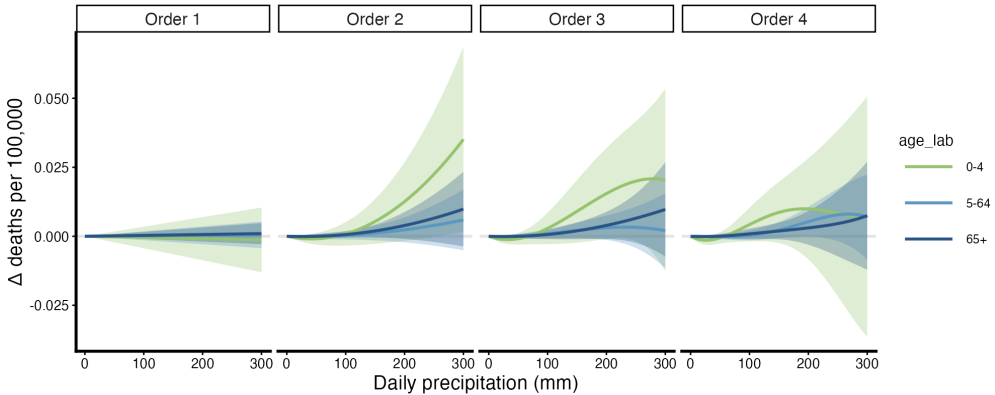
Computed under specification (1), whose year × age effects are the mildest time controls. Temperature in the Philippines is spatially smooth and dominated by national interannual (ENSO) swings, so year effects remove almost all of it: the identifying variation left in daily temperature is **0.127°C**. A quadratic over 15–35°C cannot be fitted off that. Rainfall retains roughly five times as much.



Temperature response by fixed-effects set ($K_T = 3$). The heat slope flips sign across the sets — at 33°C the 65+ response is −0.003 under year × age effects but +0.002 with province-age trends — with no stable shape. The precipitation response at 200 mm/day, by contrast, stays positive for every age group across all four specifications.

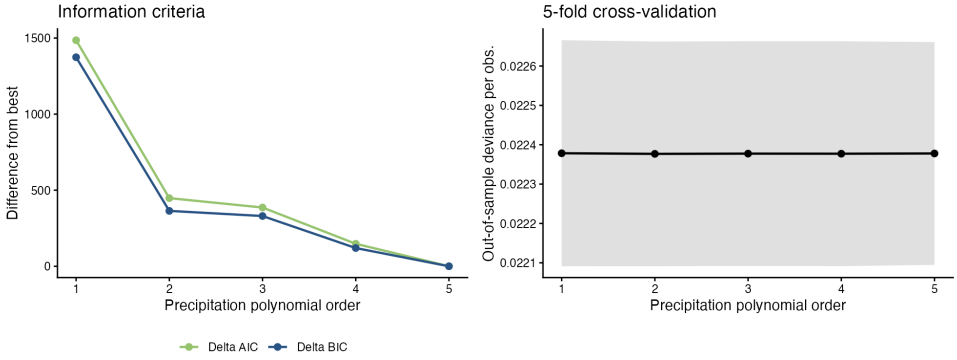
Robustness

Precipitation polynomial order



Temperature polynomial held at order 3 throughout. Order 1 forces monotonicity in daily rainfall and is shown for reference; see the polynomial-selection results for formal comparison.

Which polynomial order? Formal selection



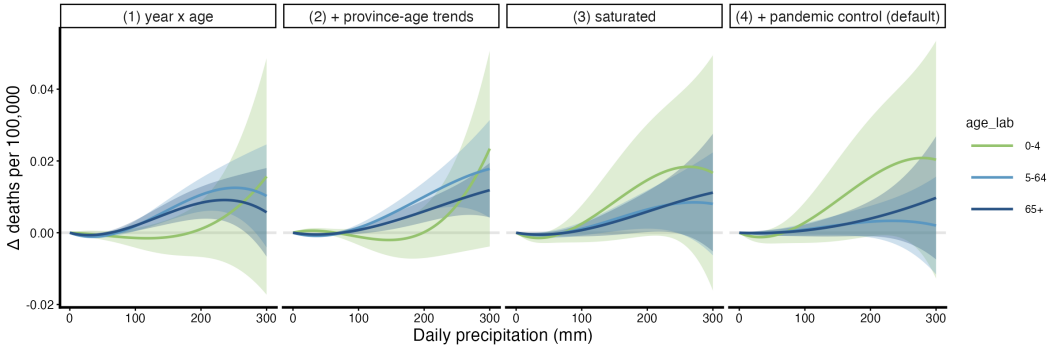
The two criteria disagree, and the disagreement is the finding. In-sample AIC/BIC fall monotonically in the order (best at 5) — the usual in-sample reward for flexibility, which at $n = 84{,}075$ the parameter penalty never offsets. But **out-of-sample deviance is flat**: the spread across orders 1–5 is 1.5×10^{-6} against a fold-to-fold standard error of 2.9×10^{-4}, roughly **190 times larger**. None of the extra curvature generalises. Order 3 (the default) is defensible; so is order 1. Computed by `03_phl_poly_selection.R`, which is kept out of the main pipeline because it refits the model 30 times.

Two caveats on the selection exercise

- **The criteria shown are from an *unweighted* refit.** Under the estimation default the likelihood is on the scale of population *shares*, so the total quasi-log-likelihood is about -751 across 84k rows and AIC's $2k$ penalty swamps it: the gap between orders 1 and 2 is 5.85 against a pure penalty of $2 \times 3 = 6$. Weighted AIC/BIC here are parameter-counting, not model comparison, and would trivially select order 1. Both variants are in `poly_selection.csv`.
- **Folds are random at the observation level**, not blocked by municipality or year — blocking would leave held-out rows with no identified fixed effect to predict from. So this measures functional form, not forecasting skill into new places or years. About 82% of held-out rows are predictable; the rest fall in fixed-effect cells that vanish from the training fold.

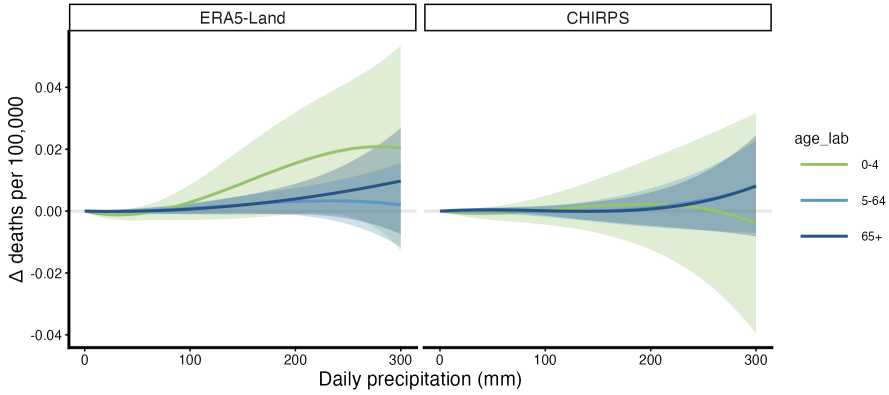
Taken with the wide confidence bands throughout, the reading is that this panel pins down the *level* of the rainfall–mortality relationship far better than its *shape*.

Fixed-effects specification



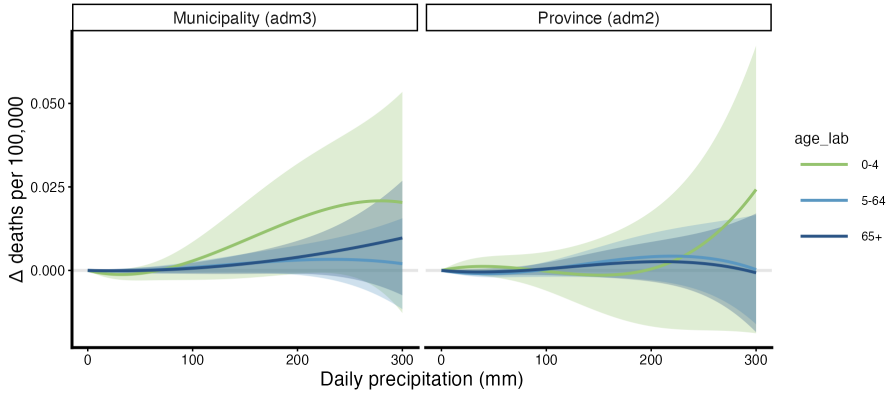
Specifications (1)–(4) as defined above.

Climate product: ERA5-Land vs CHIRPS



Both columns use ERA5-Land temperature controls; only the precipitation product differs.

Unit of analysis: municipality vs province



Left: the default municipality (adm3) panel. Right: collapsed to provinces (adm2), 6,270 observations, with fixed effects shifted up one level (province × age × era and region × year × age). Point estimates are close — at 200 mm/day and 65+, 0.0039 against 0.0026 — so the result is not an artefact of the spatial unit. Province aggregation actually *retains* slightly more identifying rainfall variation after fixed effects (23.2% vs 17.8%), because the coarser fixed effects absorb less.

How the aggregation is done

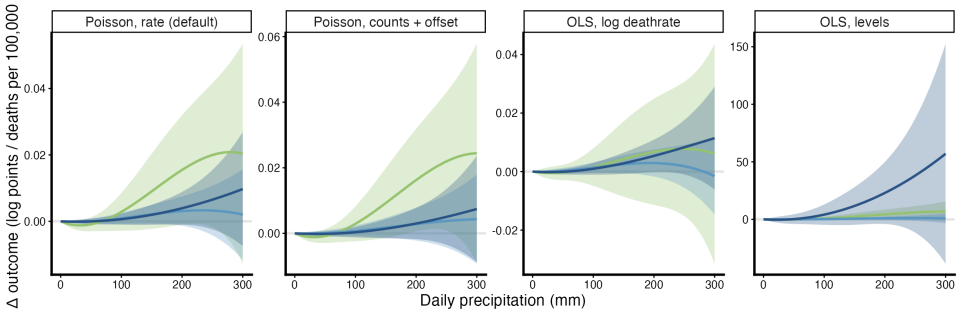
Two choices matter when collapsing municipalities into provinces:

- **The outcome aggregates exactly.** A province death rate is total deaths over total population, which *is* the population-weighted mean of its municipality rates — no modelling choice. Winsorisation is re-applied at the province level, since the 99th percentile of a different distribution is a different number.
- **The climate regressors are averaged as polynomial terms**, not recomputed from averaged rainfall:

  $$\sum_i w_i \Big(\sum_d P_{id}^k \Big) \quad \text{not} \quad \Big(\sum_i w_i \sum_d P_{id} \Big)^k$$

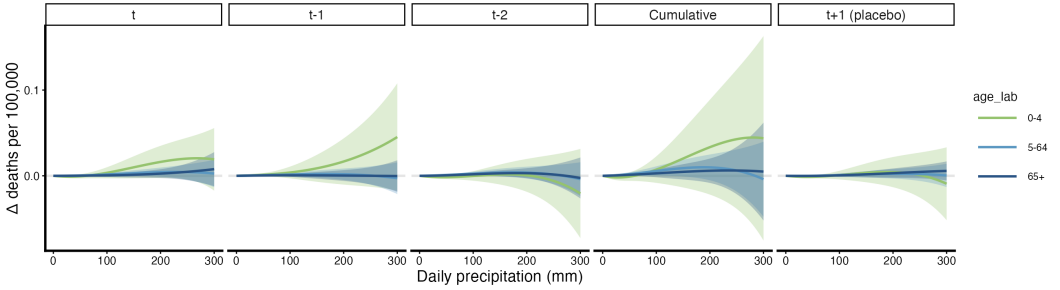
  Because the model is linear in these terms, the first makes the province response function exactly the population-weighted average of its municipalities' response functions. The second is a different object — a function of average rainfall rather than the average of the function — and would understate extremes by Jensen's inequality.
- Weights are **time-invariant** (period-mean municipal population), so that a mortality shock which moves population cannot also move measured climate exposure.

Estimator: Poisson (rate and count) vs OLS (levels and logs)



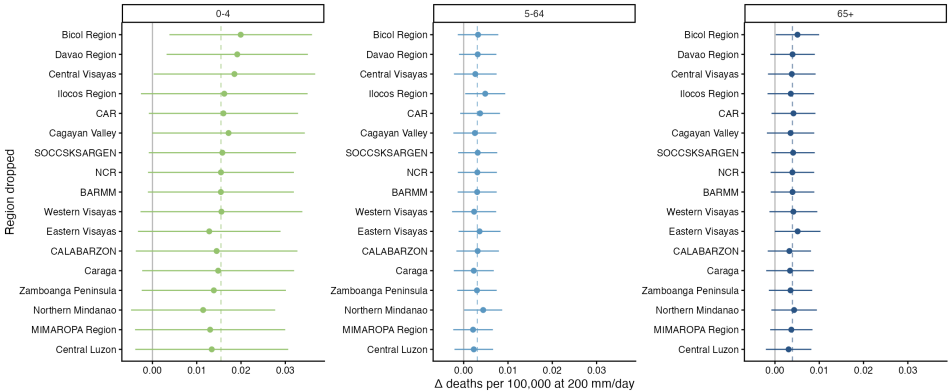
Default specification (4). The default is Poisson on the winsorised rate, population-weighted; the count model is Poisson on raw deaths with a log(population) offset, unweighted, and keeps zero-death cells. The first three panels are all in log points and directly comparable; OLS in levels (right) is in deaths per 100,000 — note the free y scales. The log-OLS sample drops zero-deathrate cells, mostly ages 0–4 ($\sim$5% of that group).

Dynamics: lags, cumulative effect, and a lead placebo



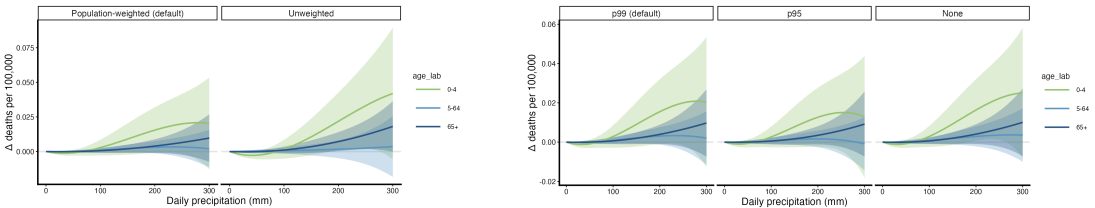
Default specification (4) with contemporaneous, one- and two-year-lagged precipitation polynomials entered jointly (first four panels; sample starts two years later to accommodate the lags). **No harvesting**: the lags enter weakly positive, not negative, and the cumulative response exceeds the contemporaneous one (65+ at 200 mm/day: 0.0059 vs 0.0026), though the cumulative CI spans zero. The rightmost panel is a falsification test from a separate regression; annual rainfall is serially uncorrelated within municipality (AR(1) ≈ -0.04), so the lead cannot proxy current exposure. Its CI spans zero at every age, but the test is low-powered — the placebo CI is wide enough to contain effects the size of the main estimate.

Leave-one-region-out



Response at 200 mm/day, re-estimated 17 times dropping one region (adm1) at a time. Dashed line marks the full-sample estimate. A result driven by a single region — e.g. the typhoon corridor — would show one point far from the dashed line.

Weights and winsorisation



Left: population-weighted (default) vs unweighted. The gap is real: at 200 mm/day the unweighted 65+ estimate roughly doubles (0.0039 $\rightarrow$ 0.0072) while 5–64 halves (0.0031 $\rightarrow$ 0.0017), pointing to larger old-age effects in smaller municipalities — consistent with the rural/urban heterogeneity section. Right: winsorisation at p99 (default, inherited from the Carleton pipeline), p95, and none — estimates barely move.

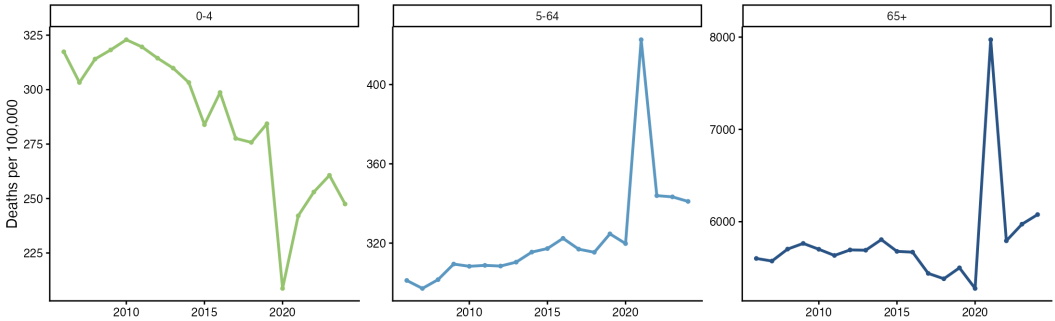
Clustering

	0–4	5–64	65+
Point estimate	0.0155	0.0031	0.0039
Province (110 clusters, default)	(0.0084)	(0.0023)	(0.0025)
Region (17 clusters)	(0.0088)	(0.0028)	(0.0021)
Two-way: province and year	(0.0126)	(0.0047)	(0.0035)

Standard errors of the response at 200 mm/day under alternative clustering; point estimates identical by construction. Region-level clustering (17 clusters) barely moves the SEs but its asymptotics are poor. **Two-way province-and-year clustering raises the SEs by 40–100%**, and at 200 mm/day no age group then excludes zero — serial or common shocks within years matter for inference here.

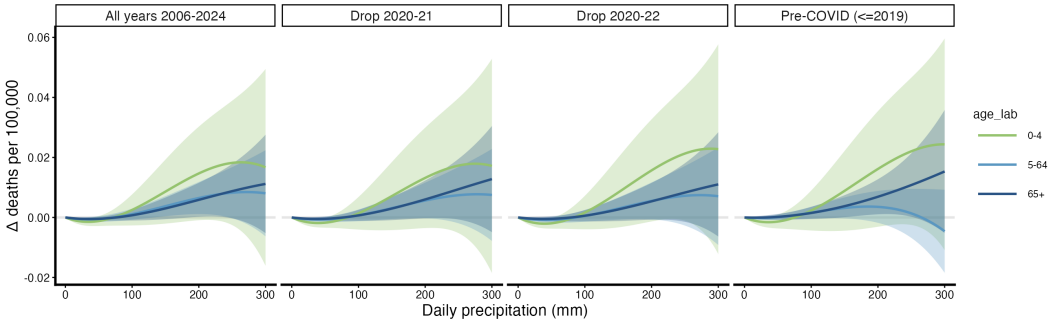
COVID

The pandemic sits inside the estimation window



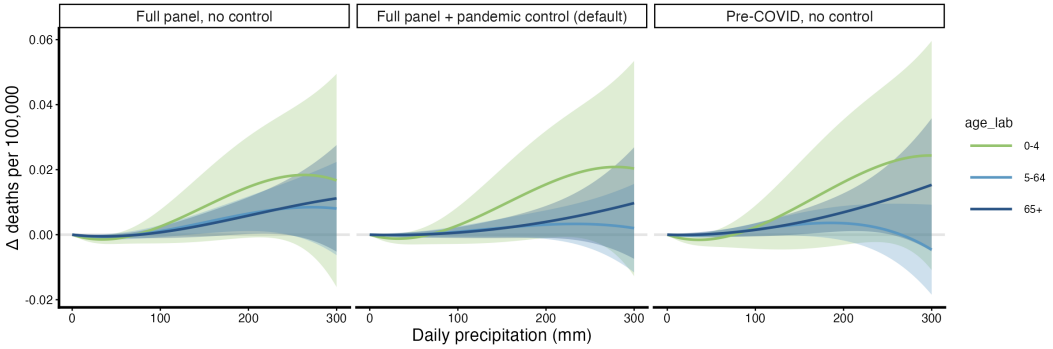
Population-weighted mean death rate by year. Mortality at 65+ rises about 45% in 2021 and does not return to its pre-pandemic level. Year $\times$ age effects absorb the national component of this, but not any covariance between local pandemic severity and local weather.

COVID sample sensitivity is modest under the saturated FEs



All four samples estimated under the saturated specification (3), so the comparison is not mediated by the pandemic control. The samples differ by well under one standard error — at 200 mm/day and age 65+, 0.0059, 0.0060, 0.0054 and 0.0069 against a standard error of about 0.003. The province $\times$ year $\times$ age effects already absorb most of the pandemic shock. Under the weaker year $\times$ age effects the same comparison showed the full sample roughly doubling the effect, so how much COVID matters is itself a function of the fixed effects.

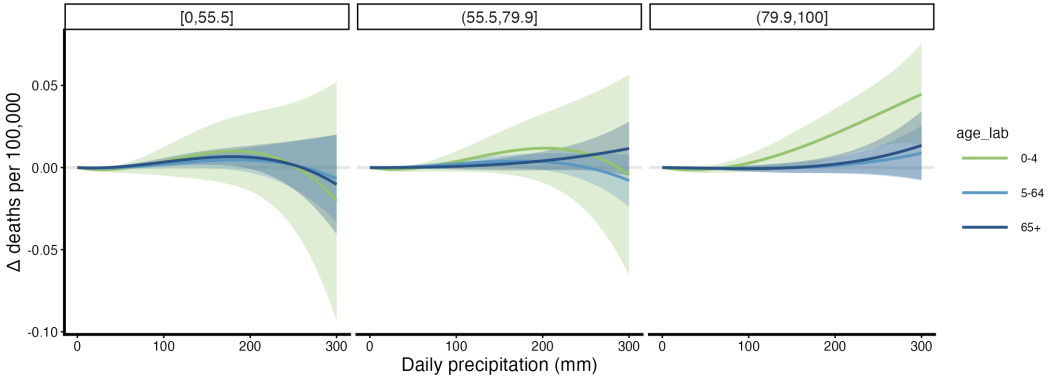
The default: a pandemic control, not a truncated sample



The saturated specification already absorbs any pandemic shock common to a province × year × age cell, so the useful control is one level lower: a separate intercept for each **municipality × age in each pandemic era** (pre-2020, 2020–22, 2023–24), which soaks up locality-specific COVID mortality while leaving within-era weather variation intact. At 200 mm/day for ages 5–64 it returns 0.0031, against 0.0035 on the pre-COVID sample and 0.0065 with no control — i.e. it recovers essentially the pre-COVID estimate while using all 19 years, with standard errors about 15% tighter — which is why this is the default here, rather than discarding five years of data.

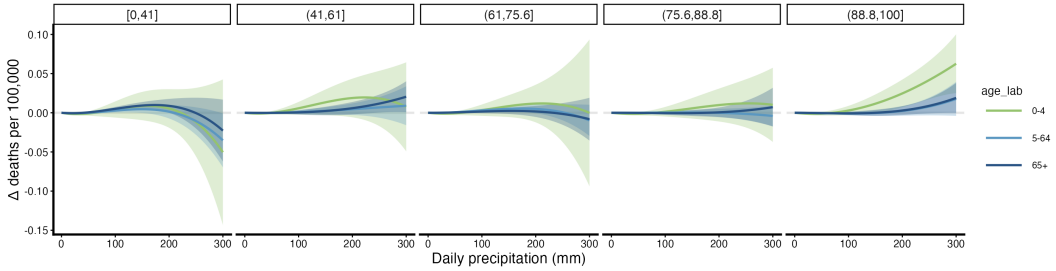
Heterogeneity

Urban population share



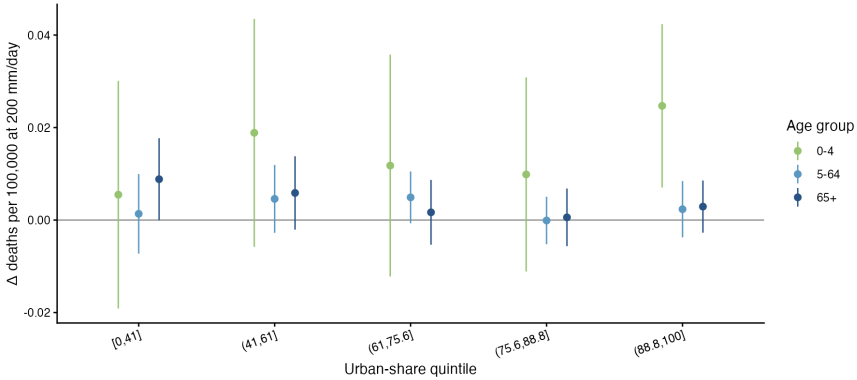
Precipitation polynomial interacted with age × urban-share bin. Bins are **terciles of the urban-share distribution**, not equal-width: urban share is left-skewed (median 69%), so equal-width bins would put 3.4 times as many municipality-years in the top bin as the bottom and estimate the rural response off far less data. Each bin here holds about 9,600 observations.

Urban population share — five bins



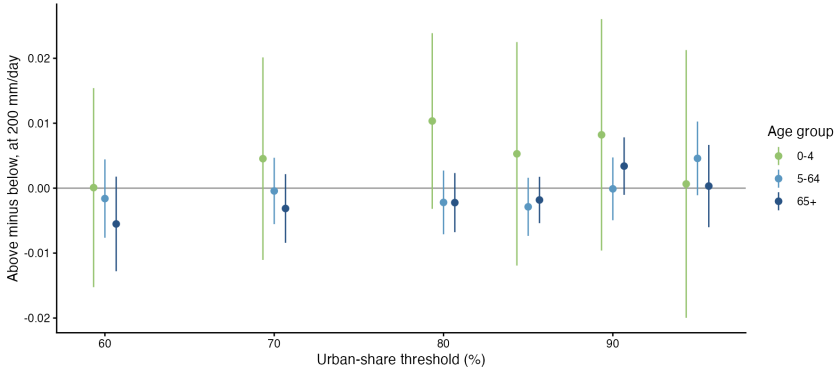
Quintiles of the urban-share distribution, about 5,800 municipality-years each. The finer grid buys little: no monotone gradient in urbanisation emerges, and only the most-urban bin for ages 0–4 excludes zero at 200 mm/day (0.025 [0.007, 0.042]). Splitting further mostly costs precision.

Is the urban gradient real? Three tests



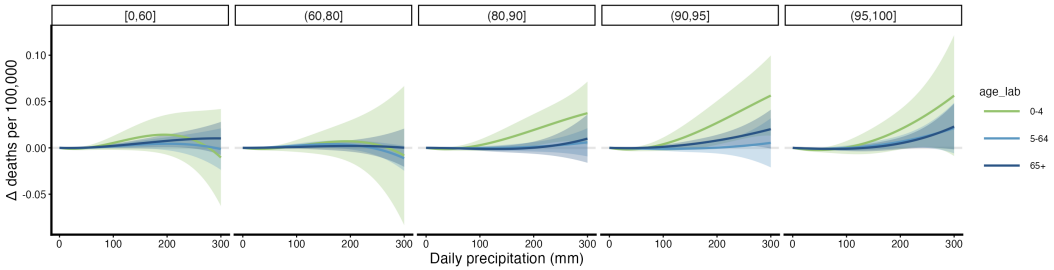
Response at 200 mm/day by urban quintile. The eye is drawn to the top-right point; the tests below ask whether that is warranted.

Testing the threshold hypothesis directly



Difference in the 200 mm/day response between municipalities above and below each urban-share cutoff. **No cutoff yields a significant difference at any age** — the smallest p-value across all 18 tests is 0.11. Testing the top quintile against *all* others pooled (the most powerful version of "only the really urban places") gives $+0.013$ ($p = 0.13$) for ages 0–4, $+0.0005$ ($p = 0.88$) for 5–64 and $+0.0002$ ($p = 0.95$) for 65+.

Splitting the top of the distribution finer



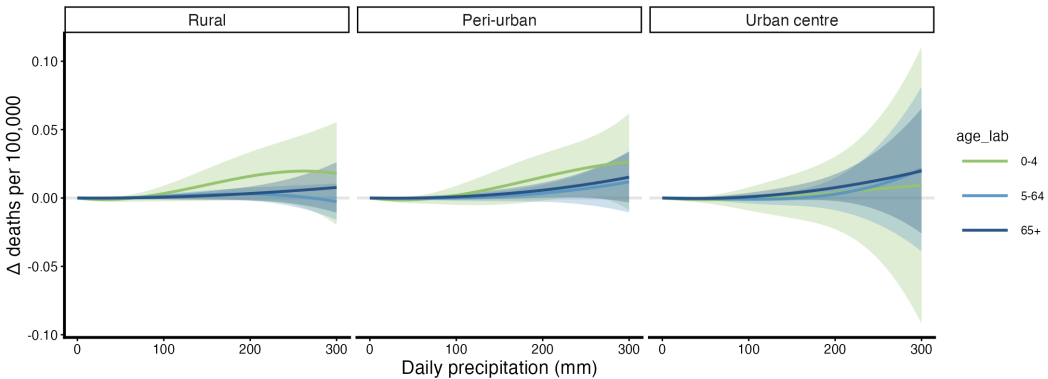
Bins chosen to be fine where the hypothesis lives: 0–60, 60–80, 80–90, 90–95, 95–100% urban. At 200 mm/day the significant cells are (80–90] and (90–95] for ages 0–4, (90–95] for 65+ — and **[0–60], the *least* urban bin, for 65+**. The **most** urban bin (95–100%) is not significant at any age.

What the urban result actually is

- **The threshold hypothesis is not supported.** No cutoff from 60% to 95% separates damaging from non-damaging municipalities, and the top quintile is indistinguishable from the rest.
- **Which bin is "significant" moves with the binning.** With quintiles it was (88.8–100]; with the finer grid it is (90–95], while (95–100] is not; and for 65+ the *least* urban bin also clears. That instability is the signature of noise rather than a threshold.
- **But there is something — and it is about age, not urbanisation.** For ages 0–4 the point estimate is positive in *every* bin (0.014, 0.007, 0.019, 0.027, 0.020) and the pooled estimate sits just short of significance (0.015, s.e. 0.008). The most defensible reading is a broad infant response to extreme rainfall across the whole urban distribution, which bin-by-bin testing makes *look* like an urban threshold.

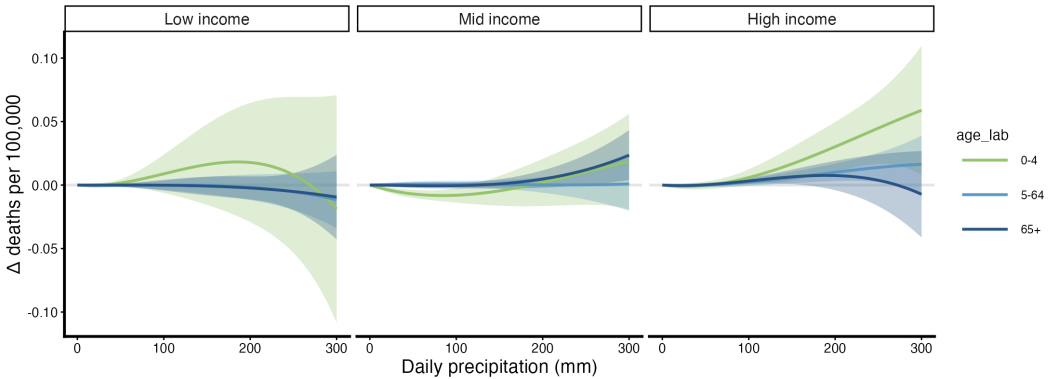
Distinguishing these needs a mechanism, not more bins: cause-of-death for 0–4 (waterborne disease, drowning) is the test, and the data is already cleaned on the cluster.

Settlement type (GHSL degree of urbanisation)



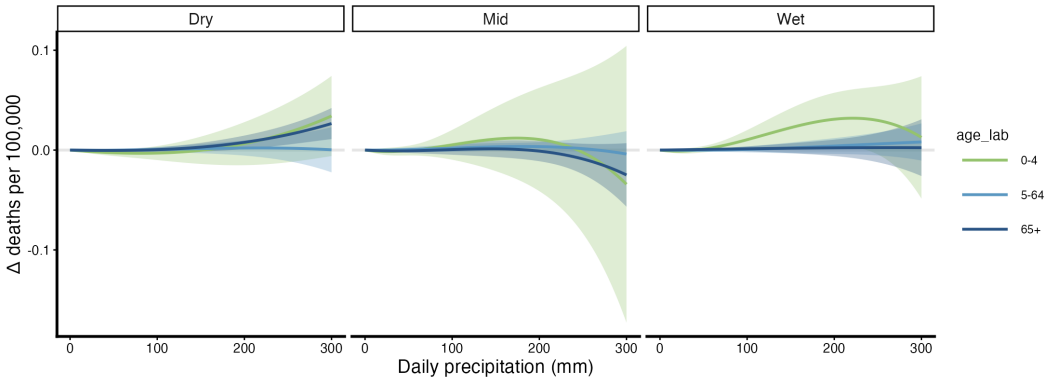
GHSL SMOD classes collapsed to rural (10–13), peri-urban (21–22) and urban centre (23, 30).

Income



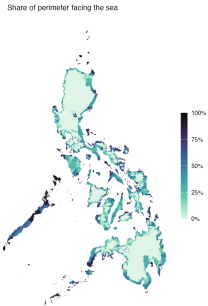
Terciles of region-level average GDP per capita. Income varies only across the 17 regions, so this margin is coarse.

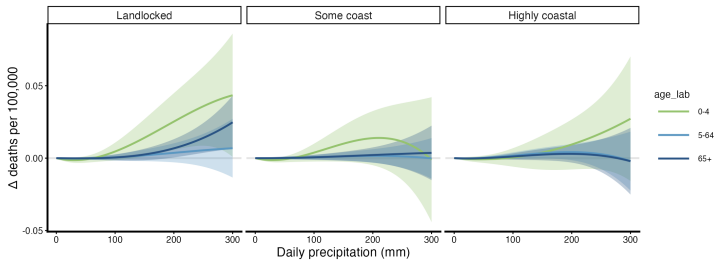
Long-run precipitation (adaptation margin)



Terciles of region-level long-run mean annual precipitation.

Coastal exposure



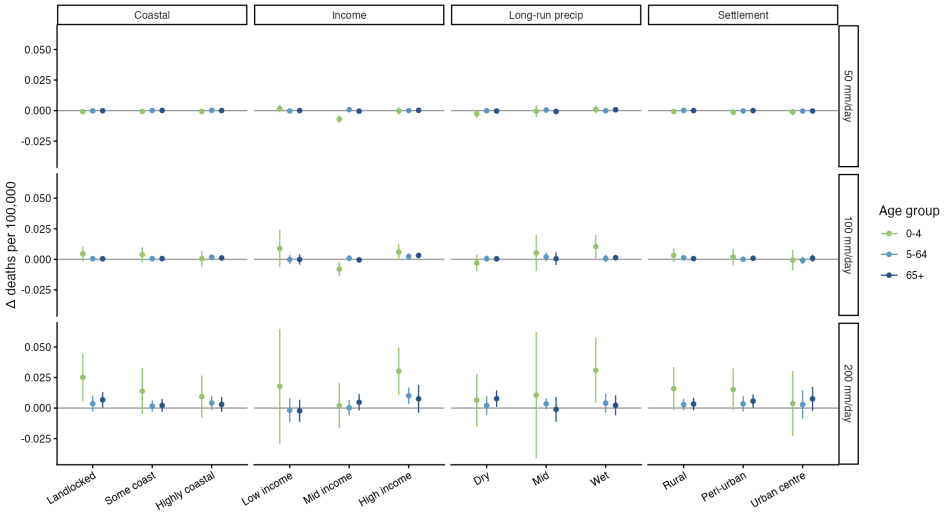


Exposure is the share of each municipality's perimeter facing the sea, computed from the adm3 shapefile by subtracting shared borders from the perimeter (`04_phl_coastal.R`). 41% of municipalities are landlocked; the coastal ones are split at their median (37%). Validation: the measure returns 0 for inland Ilocos Norte and 1 for Batanes and the Babuyan and Polillo islands.

Coastal exposure: the sign is the wrong way round

- At 200 mm/day the **landlocked** bin carries the largest effects and is the *only* one whose confidence intervals exclude zero — ages 0–4: 0.025 [0.006, 0.045]; 65+: 0.0068 [0.0005, 0.0131]. Highly-coastal municipalities show smaller, insignificant responses at every age.
- That is the opposite of a storm-surge story, and consistent with two non-exclusive readings: heavy rain drains quickly to the sea on the coast but **accumulates in inland river valleys and basins**, and coastal municipalities facing chronic typhoon exposure are **better adapted** (warning systems, evacuation, building practice) than inland ones that see extreme rainfall rarely.
- **Do not over-read it.** Only 2 of 9 age $\times$ bin cells exclude zero, the intervals are wide, and the bins are cross- sectional so the contrast is between places rather than within them. A convincing version needs the daily extreme-rainfall measures and the cause-of-death split — drowning and injury deaths should load on this margin if the flooding mechanism is right.

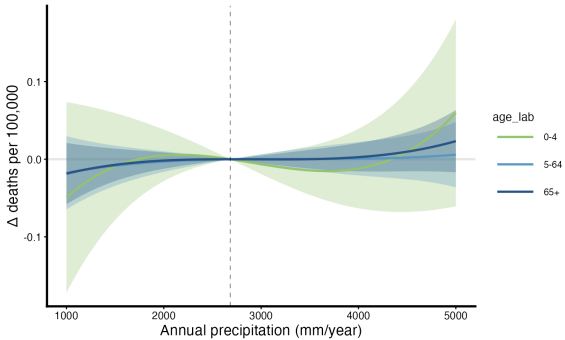
Marginal effects at 50, 100 and 200 mm/day



Point estimates with 95% confidence intervals, read off the fitted response curves at the indicated daily rainfall levels.

Alternative Functional Form

Polynomial in annual, rather than daily, rainfall



Everything else in this deck uses the Carleton form, where the order-k regressor is $\sum_d P_{id}^k$ and the curve is read over *daily* rainfall. Here the regressor is $\left(\sum_d P_{id}\right)^k$ instead, so the curve is read over *annual* rainfall.

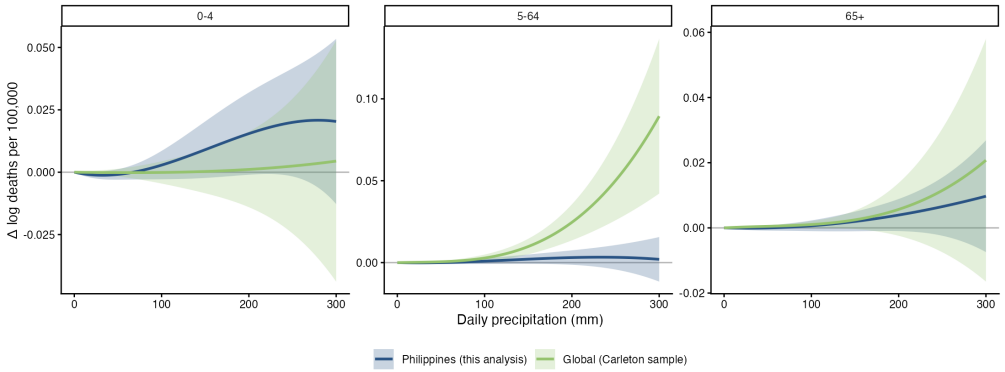
These answer different questions — what one wet day does, versus what a wet year does — and coincide only if rainfall were constant within the year.

Under the default fixed effects the annual form is very imprecise: every confidence interval spans zero across 1,500–4,000 mm/year. Province × year effects leave little annual-total variation to work with.

Dashed line marks the reference, the sample median annual rainfall. Default specification (4), full panel.

Comparison with Global Estimates

Philippines vs the project's global response



The PHL response (default specification (4)) overlaid on the global precipitation response from the project's Carleton-sample regressions (ERA5-Land, main global specification: adm2 × age and country × year × age effects). Both are Poisson, so both curves are in log points. At 200 mm/day the PHL 5–64 response (0.0031) is roughly an eighth of the global estimate (0.0246); 65+ is comparable (0.0039 vs 0.0057). Samples, admin units and fixed effects differ, so read this as context rather than a like-for-like test.

Drought vs Extreme Rainfall (Prototype)

Separating drought from extreme rainfall: design

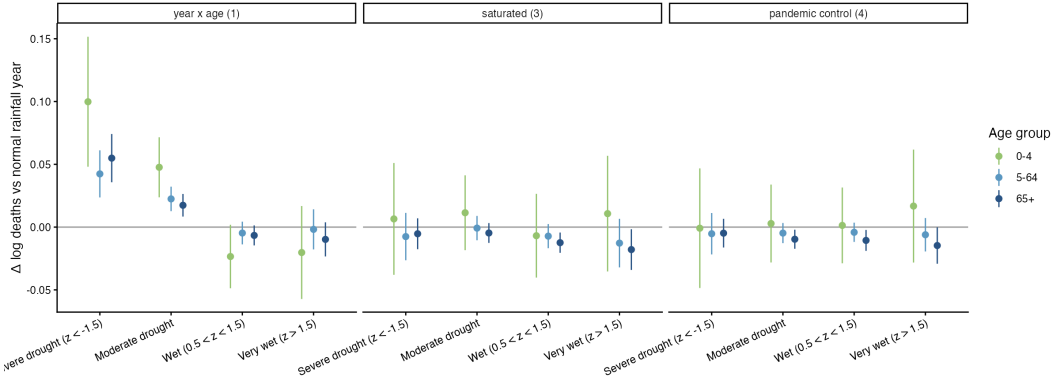
Drought and flood mortality operate at different timescales, so they enter as *separate exposures* in one regression:

$$D_{ait} = \exp\Big(\underbrace{f_a(\text{daily polynomial})}_{\text{wet-day extremes}} + \underbrace{g_a(\text{annual anomaly bins})}_{\text{cumulative deficit/surplus}} + \text{temp} + \mu_{aic(t)} + \nu_{atr(i)} \Big) \cdot \varepsilon_{ait}$$

- Prototype drought index: within-municipality **z-score of annual rainfall** (a poor man's SPI — monthly totals for a real SPI need the cluster climate build). Bins: severe ($z < -1.5$), moderate (-1.5 to -0.5), normal (reference), wet, very wet.
- **Identification checked first** (table below): the absorption is substantial — the saturated effects remove three-quarters of the z-score's variation — but far from fatal: the severe-drought *dummy* keeps about half its variation, versus 9% for temperature. PHL droughts have more subnational texture than the ENSO narrative suggests.
- Bin shares: severe drought 6.9% of municipality-years, moderate 25%; symmetric on the wet side.

Fixed effects	Annual z-score	Severe-drought dummy
	% of s.d. remaining	
year x age (1)	73.3	81.5
saturated (3)	29.0	52.3
pandemic control (4)	25.9	50.9

Prototype results: annual-anomaly bins, by fixed effects



Effect of a drought/wet year relative to a normal rainfall year ($|z| \leq 0.5$), estimated alongside the daily precipitation polynomial and temperature controls. Under the weak year $\times$ age effects, drought years show *higher* mortality (65+ moderate drought: +5.5%), but this collapses — and for 65+ turns significantly *negative* (-1.0%, se 0.4%) — once province $\times$ year effects are added, the signature of regional confounding rather than a drought effect. The precise near-zero under the default is itself informative: in this wet, non-agrarian-margin country there is no detectable drought mortality. The multi-country panel, with drought-prone regions and staggered timing, is where this margin gets its real test.

Caveats

Caveats and next steps

- **The pandemic control is doing real work only for 5–64.** It reproduces the pre-COVID estimate closely for ages 5–64, but for 65+ it lands below both the uncontrolled and the pre-COVID estimate. Under the saturated fixed effects the COVID-sample sensitivity is in any case well within one standard error — it was large only under the weaker year × age effects.
- **The temperature response is not identified.** Every age group is wrong-signed somewhere — hotter days imply lower mortality for 5–64 and 65+, colder days imply lower mortality for 0–4 — and the shapes reshape substantially across fixed-effects sets and polynomial orders. After the fixed effects only 0.127°C of daily temperature variation remains and the polynomial terms are 0.99 collinear (VIF 55), so the curves are essentially extrapolation. Temperature is kept as a control and should not be interpreted.
- **Income variation is coarse.** `gdppc_adm1_avg` takes only seven distinct values across 17 regions, so the income margin is close to a regional fixed effect.
- **Clustering.** Standard errors are clustered at the province (adm2, 110 clusters); the default fixed effects are also at province × year × age, so the identifying variation is within-province. Two-way province-and-year clustering raises the SEs by 40–100% and removes what significance there is at 200 mm/day — the inference is fragile to common within-year shocks.
- **Coverage.** Roughly 3% of municipality-years are missing ERA5-Land climate and drop